

ME2 - VCM on Raspberry Pi 5

Two-stage, from-scratch voice command model

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Abstract

This report documents the current training artifacts and code, including measured limits. A binary 2-class wake detector continuously evaluates local audio, then a 31-class intent classifier maps an accepted command to a coded action. Both use the same TinyDSCNN-48 topology and a 40-band log-mel frontend. Weather uses a separate live API; recognition itself uses no cloud model. The staged INT8 pair totals **76740 bytes**. The 95% per-class goal is assessed on a reused speaker-held-out command set and is not assumed to be met.

1 Requirements and command mapping

The course asks for common smart-device interactions, multi-speaker speech, scratch training, a tiny offline classifier, and a Raspberry Pi demonstration. The class dataset contains 17986 command WAV rows from 100 reference speaker IDs. The 31 output labels refine the ten requested command families into fixed slot values. The wake phrase is a separate binary output, not a 32nd intent class.

Course command family	Dataset labels
Play music	PLAY_MUSIC
Questions: weather and time	WEATHER, TIME
Lights on and off	LIGHT_ON, LIGHT_OFF
Brightness and color	BRIGHTNESS_20, BRIGHTNESS_60, BRIGHTNESS_100, COLOR_RED, COLOR_GREEN, COLOR_BLUE
Timers	TIMER_10s, TIMER_30s, TIMER_1m
Alarms	ALARM_6_00AM, ALARM_8_00AM, ALARM_9_00PM
Thermostat setpoints	TEMPERATURE_18, TEMPERATURE_22, TEMPERATURE_26
Media and volume	PAUSE, STOP, NEXT, VOLUME_UP, VOLUME_DOWN
Reminders and lists	LIST_REMINDERS, CREATE_REMINDER_DRINK_WATER, CREATE_REMINDER_EXERCISE, CREATE_REMINDER_STUDY
Calls and messaging	CALL, MESSAGE

The recorder's 31 intent choices equal the model label order and manifest label set. Each fixed suggested phrase appears in the matching class's training transcripts, so the suggested prompts were retained rather than changed after collection. Four additional recorder choices are `wake_word`, `_unknown_`, `_background_noise_`, and `_silence_`. Exact legacy command mappings are accepted; unsupported old slot values stay excluded. The machine-readable check is `docs/ME2_DATASET_ALIGNMENT.json`.

2 Data provenance and split discipline

The dataset's local README attributes its reference voices to LibriSpeech and SilencioPH. The manifest uses 14370 train, 1818 validation, and 1798 test rows, with reference speakers separated by partition. WAVs are read and resampled to 16 kHz by the frontend when necessary. The human recorder now holds 851 manifest rows from three anonymous IDs, dominated by person-01; exact

waveform deduplication leaves 614 unique PCM groups. All 31 intent classes have some mapped personal examples, but amounts and acoustic conditions remain uneven. One exact waveform appears under both `CREATE_REMINDER_STUDY` and `MESSAGE`; both original files remain and the ambiguous audio is excluded from training/evaluation. The personal split is by deduplicated waveform within label, so speakers can occur in training and test. These personal scores are *not* unseen-speaker evidence.

The wake optimizer labels every command in the reference *training* partition as `NON_WAKE`, along with training-only personal commands, unrelated speech, room noise and silence. The positive class is the recorded wake phrase. Validation and test partitions are not optimization inputs. Class weighting counters the large negative-to-positive imbalance. The intent optimizer uses the 31-class reference training partition and exact-mapped personal training takes. No pretrained weights or ASR model are loaded.

3 Signal path and architecture

Capture is mono at 16 kHz. A 2.5 s window contains 40,000 samples. The frontend computes a 512-point short-time Fourier transform using a 400-sample Hann window and 160-sample hop, projects power to 40 mel bands, applies log compression, then standardizes the feature tensor. The resulting network input is (1, 40, 251) per item. The 25 ms window, 10 ms hop, and FFT size are distinct quantities; the 512-point transform zero-pads each 400-sample analysis frame.

The stem is a 5×5 convolution with stride (2, 2), 48 output channels, batch normalization and ReLU. Four depthwise-separable blocks each contain a 3×3 channel-wise convolution, batch normalization, ReLU, a 1×1 pointwise convolution, batch normalization and ReLU. Block 2 halves the time dimension. Global average pooling yields 48 features, followed by dropout $p = 0.15$ during training and a linear head. Only that head differs between the two models.

Stage	Output (C, F, T) or vector	Parameters
Input log-mel	(1, 40, 251)	0
5x5 stem / BN / ReLU	(48, 20, 126)	1,296
DS block 1	(48, 20, 126)	2,928
DS block 2	(48, 20, 63)	2,928
DS block 3	(48, 20, 63)	2,928
DS block 4	(48, 20, 63)	2,928
Global average pool	(48,)	0
Linear head: 31 logits	(31,)	1,519
Intent total	31 logits	14,527
Wake total	2 logits	13,106

For 48 input and output channels, a conventional 3×3 convolution has $9 \cdot 48^2 = 20,736$ kernel weights. Each depthwise/pointwise pair has $9 \cdot 48 + 48^2 = 2,736$ kernel weights, before normalization parameters: about 7.58 times fewer convolution weights. The full heads contain 13,106 and 14,527 parameters. Static QDQ INT8 ONNX export uses training-only calibration features.

4 Training and measurements

Both models were randomly initialized and trained in the shared PyTorch environment. The latest run is `me2-vcm-20260930-human851`: binary wake used seed 231 for 30 epochs; refined 31-class

Always-listening two-model speech-to-action path

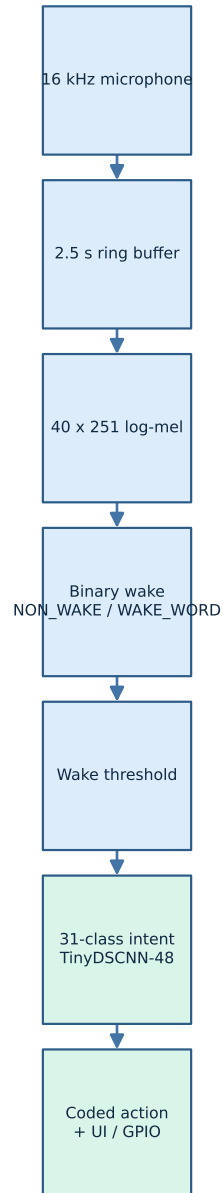


Figure 1: The actual two-model routing. The wake gate precedes intent inference; actions are coded, not generated by an LLM.

TinyDSCNN-48 layer connections (31-class intent head)

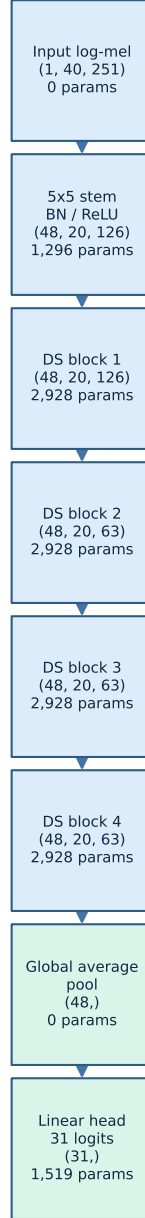


Figure 2: Layer-stage connections drawn with NetworkX from a live PyTorch shape and parameter inspection.

intent used seed 232 for 60 epochs. The intent optimizer mixed 315 deduplicated personal training clips with the 14370 reference training clips. AdamW with cosine learning-rate decay optimized both models; time/frequency masks and limited temporal jitter augmented training. Checkpoints and wake calibration used validation data. The reference test partition was used in earlier iterations, so its score is a reused benchmark rather than a pristine one-time test.

At the fixed wake operating threshold of 0.95, saved-clip replay accepted **8/8** held-out personal wake clips, falsely accepted **0/120** personal non-wake clips, and falsely accepted **0/1798** reference commands. This is a finite saved-clip replay, not continuous-room false-activation-per-hour testing. The validation-selected threshold was 0.9996; 0.95 is the fixed operating choice. The wake set remains small.

The INT8 intent classifier reached **97.22% accuracy** and **97.22% macro F1** on 1798 reused reference command test clips. On 101 held-out personal takes it reached **92.08% accuracy** and **93.62% macro F1**; speaker IDs overlap training. Reference classes below 95% F1 are blue color, red color, lights off, reminder list, and stop. Personal classes below 95% F1 are the 8:00 alarm, calls, lights off, next, pause, play music, and time. The target of at least 95% F1 per class is not met. Personal evaluation is especially uncertain for classes with only a few held-out takes.

Intent class	Ref F1	Recall	n	Personal F1	n
ALARM_6_00AM	99.1	98.3	58	100.0	2
ALARM_8_00AM	99.1	100.0	56	80.0	2
ALARM_9_00PM	100.0	100.0	58	100.0	3
BRIGHTNESS_100	96.8	100.0	60	100.0	4
BRIGHTNESS_20	97.5	98.3	60	100.0	3
BRIGHTNESS_60	98.3	96.7	60	100.0	3
CALL	97.2	96.3	54	66.7	2
COLOR_BLUE	94.0	91.7	60	100.0	3
COLOR_GREEN	98.3	96.7	60	100.0	3
COLOR_RED	94.5	100.0	60	100.0	3
CREATE_REMINDER_DRINK_WATER	97.6	100.0	60	100.0	3
CREATE_REMINDER_EXERCISE	99.2	98.3	60	100.0	3
CREATE_REMINDER_STUDY	99.2	98.3	60	100.0	3
LIGHT_OFF	94.3	96.7	60	80.0	5
LIGHT_ON	95.2	92.6	54	100.0	4
LIST_REMINDERS	94.7	93.1	58	100.0	2
MESSAGE	97.3	94.8	58	100.0	3
NEXT	99.2	98.3	60	66.7	2
PAUSE	98.2	98.2	56	61.5	9
PLAY_MUSIC	98.3	100.0	58	61.5	4
STOP	94.5	96.3	54	100.0	7
TEMPERATURE_18	95.7	93.3	60	100.0	3
TEMPERATURE_22	96.0	100.0	60	100.0	4
TEMPERATURE_26	98.3	96.7	60	100.0	3
TIME	95.1	90.7	54	85.7	3
TIMER_10s	95.0	95.0	60	100.0	3
TIMER_1m	99.1	100.0	56	100.0	3
TIMER_30s	96.6	95.0	60	100.0	3
VOLUME_DOWN	98.1	100.0	52	100.0	3
VOLUME_UP	99.2	98.3	60	100.0	1
WEATHER	98.1	100.0	52	100.0	2

5 Runtime, controls, and hardware

In standby, only the binary wake model runs after an energy gate. After wake, the intent model runs on the utterance. The ten-second command deadline now expires even if unrelated sound or pending speech persists. The Assistant and Studio both show the RGB light state. The device module implements fixed light levels/colors, playback/next/pause/stop, timer/alarm state, and temperature setpoints. A local generated melody provides deterministic music for an offline demo. Calls and messages are explicitly simulations; no telephony API is connected.

Weather is fetched asynchronously from the Open-Meteo current-weather API when networking is available; no weather API key is required. A failure returns a clear unavailable message without blocking microphone inference. Recognition remains local. The UI can change its capture microphone without reloading models; it retries after disconnects. Browser audio output selection uses `setSinkId` where supported. One cached browser speech voice is used consistently, and its volume follows the slider; the Pi also adjusts the ALSA mixer when available. A separate Restart VCM control remains for full service recovery. The recorder now exposes an immediate **Record again** control after saving.

6 Raspberry Pi deployment and remaining gates

The active Pi release is `dandan-vcm-20260930.zip`, installed under `/home/dalmacio/dandan`. It is an application package, not an SD-card image. It includes both INT8 ONNX models, hashes, the frontend, action code, local demo music, an executable manual `launch-vcm.sh`, and an on-device verifier. It does not enable boot autostart. On ARM64, frontend-plus-model p_{95} was 5.855 ms for wake and 5.335 ms for intent. The Pi PCM mixer responded to a 35% volume request and was restored to 100%. GPIO red/green/blue/off checks passed. The Pi currently has no capture device, so live speech remains untested. Sixteen prior release folders and ZIPs are archived under `/home/dalmacio/archive/legacy-releases-20260930`; the existing `/home/dalmacio/vcm` project was left untouched. Windows users can launch the Pi and open the tunneled Assistant by double-clicking `C:/Users/danda/Desktop/dandan/Start-VCM.bat`.

For a class demonstration, show the wake transition first, then light brightness/color, volume, local music, temperature, and weather. State plainly that the RGB panel and thermostat are simulations unless GPIO/HVAC hardware is connected. For a stronger performance claim, record multiple sessions and all 31 classes from additional speakers, hold complete speakers out, collect room noise and near-miss wake phrases, and measure continuous false activations per hour. Create a fresh sealed test set for the next evaluation.